Speaker-Attributed Transcription of Bengali Doctor–Patient Consultations

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AIMScribe — Technical Assessment Report · October 2026

Code, notebooks and data: https://github.com/sahilaf/Aimscribe

Submission files (Google Drive):
https://drive.google.com/drive/folders/1njMEFzB3lV_t9gzqK0UdIe3RrVoisOUT?usp=sharing

Summary. **Task 1:** on the held-out Indic DiarBench Bengali test set (9.3 h of real speech), pyannote *community-1* lowers DER from 19.8 % (pyannote 3.1) to **16.7 %**. My segment re-clustering lowers DER on simulated conversations from 28.1 to 15.9 % but does not transfer to real meetings; my overlap fusion lowers DER only *inside overlap* (38.0 → 33.7 %) and is not used. **Task 2:** on the OpenSLR 53 test set, fine-tuning whisper-small lowers WER/CER from 144.1/135.7 to **66.4/21.4 %**; a larger token budget and a VAD gate fix cut-off lines and hallucination. On my real role-play consultations (a different set) the selected IndicConformer 600M reaches 7.2 % CER. **Task 3:** the pipeline produces role-labelled transcripts. With correct speaker segments the role rules are right in 8/8 consultations; end to end they are right in every 2-speaker consultation and fail in every one with 3 or more speakers, where diarization merges people. Real-consultation results (DER 2.7 %, CER 6.8 %) come from 2.2 min of development data, not an independent test. Main limitation: clinical vocabulary.

1 Task 1: Speaker diarization

1.A Dataset preparation

Table 1: Diarization data. All decisions are made on dev data; the DiarBench test set is scored once.

Set	Content	Use
Indic DiarBench Bengali	public (CC BY 4.0), 12.6 h real meetings, far-field and in-the-wild speech, 2–8 speakers, human labels	dev 3.3 h / **test 9.3 h**
Simulated conversations	10 × 76 s built from OpenSLR 53 speakers not used for ASR; 2–3 speakers, 1–10 % overlap, short backchannels, noise on 2 clips; exact labels	development
My consultations A, B	2 real role-play recordings (2.2 min, 2 speakers); timestamps drafted automatically, corrected by ear	development

DiarBench is split by recording (clips of one recording share speakers and room), stratified by condition (`data/prepare_diarbench.py`). **Overlapping speech** is annotated in RTTM as lines from different speakers that cover the same time; the overlap regions are also written as their own RTTM files so DER can be scored inside and outside overlap. Example: the doctor interrupts at 5.27 s while the patient speaks until 5.77 s:

```
SPEAKER S05 1 2.467 3.300 <NA> <NA> Patient <NA> <NA>
SPEAKER S05 1 5.267 1.584 <NA> <NA> Doctor  <NA> <NA>
```

1.B Model implementation

Pretrained `pyannote/speaker-diarization-3.1` (`task1_diarization/diarize.py`) writes one RTTM per recording: speaker segments with start time, duration and speaker ID. DER is computed with `pyannote.metrics` at collar 0.25 s (0 s also reported in `results/`), and additionally inside and outside the reference overlap regions.

1.C Error analysis and improvement

Errors in the model output (timelines: `results/task1/baseline/plots/`, Task 1 notebook):

Table 2: DER (%) before and after each change. Simulated: tuned on even clips, reported on odd clips. Overlap fusion improves only the DER inside overlap and is not part of the final system.

						DER by region	
Data	System	DER	Miss	FA	Conf.	Overlap	Rest
Simulated, test half	pyannote 3.1	28.1	2.2	0.0	25.8	–	–
	+ segment re-clustering	**15.9**	2.2	0.0	**13.7**	–	–
DiarBench dev, 32 clips	pyannote 3.1	17.8	–	–	–	40.2	–
	community-1	**16.9**	–	–	–	38.0	–
	PixIT	42.0	–	–	–	39.4	–
	community-1 + overlap fusion (not used)	17.6	–	–	–	**33.7**	–
DiarBench test, 9.3 h	pyannote 3.1	19.8	6.4	3.1	10.3	38.2	18.0
	community-1	**16.7**	6.4	3.1	**7.3**	**35.0**	**15.0**

- **Incorrect speaker assignment:** `bn_conv_00`, 15.4–17.3 s, a turn given to the other speaker.
- **Overlapping speech:** `bn_conv_00`, 18.3–19.2 s, a 0.9 s backchannel spoken over another turn; 83 % of it is given to the main speaker.
- **One speaker identified as multiple speakers:** my consultation B, where the attendant's lines B9, B11 and B13 get a third speaker label (Figure 1).
- **Two speakers merged:** `bn_conv_01`, 2 speakers found as 1.
- **Missed short utterances:** none found in the simulated set or my consultations; short replies are detected but given to the wrong speaker.

Improvements implemented (Table 2):

1. **Clustering — segment re-clustering** (`recluster.py`). Embeddings of whole utterances separate the simulated speakers perfectly, but pyannote 3.1 embeds 10 s windows that mix two or three short turns. Re-clustering keeps pyannote's segments, embeds each whole segment (overlap removed) and clusters them again (average linkage, cosine threshold 0.70 chosen on the dev half): DER 28.1 → 15.9 %. On real meetings it does not help (DiarBench dev 18.3 → 20.8–30.5 %): turns are longer, and real overlap breaks its assumptions.
2. **Overlap handling — overlap fusion** (`overlap_fusion.py`). On real speech, DER inside overlap is 2–3× the DER elsewhere and the second voice is never detected. PixIT (joint diarization and separation) finds it (missed speech 10.2 → 3.7 %) but adds 18 % false alarm. The fusion keeps community-1 and adds a speaker only where PixIT hears overlap next to that speaker's own turn: DER inside overlap 38.0 → 33.7 %, but overall DER 16.9 → 17.6 %. An overlap-specific gain with an overall cost, so it is not used.
3. **Final choice — community-1** (newer pyannote pipeline, VBx clustering), best overall on dev and scored once on test: DER 19.8 → 16.7 % (far-field 33.3 → 24.5 %), speaker count right on 46 of 56 clips (35 before).

2 Task 2: Bengali ASR

2.A Dataset preparation

Data. OpenSLR 53 (public Bengali read speech, CC BY-SA 4.0), part 0. Speaker-disjoint split: train 80 min / 65 speakers, test 18 min / 15 speakers (`data/prepare_asr.py`). **Audio preprocessing.** Converted to 16 kHz mono WAV; long silences trimmed (0.15 s pad kept); loudness normalised. **Segmentation.** OpenSLR is already one sentence per file. My consultations are cut into one segment per spoken line using the hand-corrected timestamps (46 lines). In the pipeline, ASR segments are the diarization turns (≤ 25 s). **Transcription preparation.** Lines with digits or Latin letters are

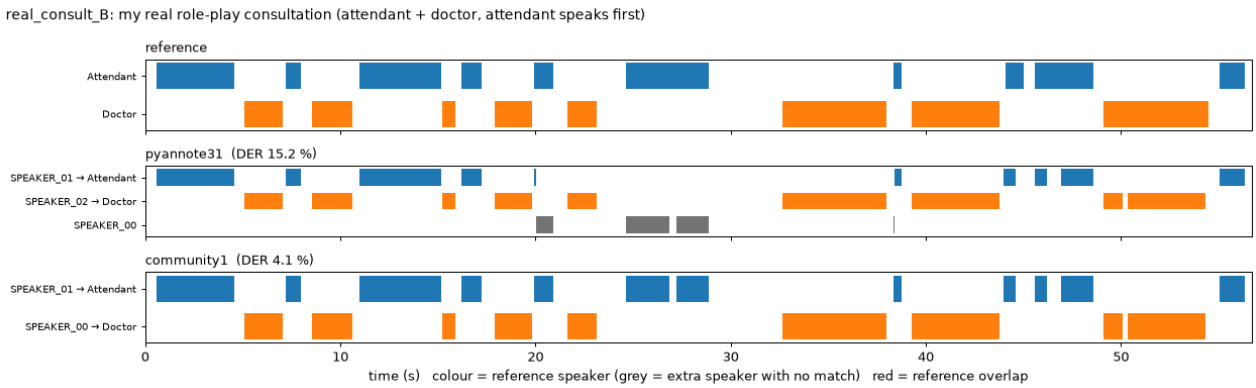


Figure 1: Sample with reference and predicted annotations: my consultation B. Pyannote 3.1 splits the attendant into two speakers (DER 15.2 %); community-1 does not (DER 4.1 %).

Table 3: Task 1 deliverables.

Deliverable	Where
Working SD implementation	`task1_diarization/` (`diarize.py`, `recluster.py`, `diarize_overlap_aware.py`, `overlap_fusion.py`, `evaluate.py`)
Sample audio with reference and predicted annotations	`data/samples/diarization/`: 9 recordings, each with audio, reference RTTM, overlap RTTM, predicted RTTM before / after, timeline
DER results	Table 2; per clip in `results/task1/<run>/der_collar0.25.csv`
Three or more actual errors	Section 1.C (five examples)
Before / after comparison	Table 2

dropped (they can be read in several ways); text is normalised (Unicode NFC; punctuation, danda and zero-width characters removed). The same normalisation is applied to every model's output before scoring.

Evaluation sets. (1) OpenSLR test: read speech. (2) My consultations: real conversational speech, 46 lines, used for model selection. (3) Synthetic clinical test: 600 TTS clips with drug, dose and test names. Their numbers are not comparable with each other; every result below names its set.

2.B Model implementation and fine-tuning

Pretrained model: `openai/whisper-small` (language Bengali, task transcribe). **Fine-tuning** (`finetune_whisper.py`, Colab T4, 34 min): all weights, learning rate 10^{-5}, batch 8×2, 800 steps, fp16; 1,481 training lines (72.5 min, 59 speakers); 6 training speakers held out as dev; best dev-WER checkpoint (step 600).

Table 4: Baseline vs fine-tuned, OpenSLR 53 test (15 unseen speakers, 390 lines).

Model	WER (%)	CER (%)	Output in wrong script
whisper-small, zero-shot	144.1	135.7	98.5 % (Devanagari)
whisper-small, fine-tuned	**66.4**	**21.4**	0 %

2.C Error investigation

Errors of the fine-tuned model on my real consultations (all six categories in the brief):

Table 5: Actual transcription errors (reference → output) and their cause.

Error type	Example	Cause
Missing words at start / end	start: হুম dropped (A5, A16, B8); end: 3 of 46 lines cut off (A21, A25, B12)	128-token limit: Whisper spends about 2 tokens per Bengali character
Short utterances	না → নাম; জি → চি; আচ্ছা → invented text	little acoustic context
Code-switching	ভাইরাল ফিভার → ওয়াই রালফিবার; সিবিসি → চিভিসি	English words rare in the training text
Medical terms	প্যারাসিটামল → পেরাসি তেমল	rare words without a fixed Bengali spelling
Hallucination	all 19 silent pauses produced text, e.g. ক্যাসায়াসায়ায় ক্যায়ায়ায়	the decoder keeps writing when there is no speech
Background noise	CER 39.2 → 44.0 / 54.4 / 61.6 % at 20 / 10 / 5 dB SNR	trained on clean read speech

2.D Practical improvement

Two problems from Table 5 were fixed at their cause, one change at a time:

Table 6: Before / after, fine-tuned whisper-small on my consultations.

Change	Problem addressed	Before	After
Token budget 128 → 440	lines cut off at the end	CER 41.6 %, 3 cut lines	**CER 39.2 %**, 0 cut lines
Silero VAD gate	hallucination on silence	19/19 pauses with text	**0/19** (speech unchanged)

Tried and rejected on the same data: 0.2 s padding, a medical-vocabulary prompt, VAD cropping.

Model selection. The fine-tuned whisper-small (39.2 % CER) is not usable, so stronger pretrained Bengali models were compared with a rule fixed in advance: lowest CER on my consultations. OpenSLR scores were not used: tugstugi whisper-medium reproduces 88 % of OpenSLR test sentences word for word (it was trained on this corpus).

Term accuracy (synthetic clinical test): 95 drug, dose and test terms (81 distinct) × 6 voices = 570 occurrences. A term counts only if the whole term begins a word of the output (Bengali suffixes allowed); no partial credit. For the clinical fine-tune, the 570 split into 228 terms seen in its training text, 222 unseen, and 120 dose words in neither group.

Table 7: ASR candidates.

		My consult.	Synthetic clinical test	
Model	Trainable	CER	CER	Term acc.
whisper-small, my fine-tune	yes	39.2	–	–
tugstugi whisper-medium	yes	13.0	3.7	57.9
IndicConformer 120M, CTC / RNN-T	yes	11.1 / 9.1	4.6	51.6
IndicConformer 120M + my clinical fine-tune (CTC)	yes	9.4	3.6	70.7
IndicConformer 600M (selected)	no (ONNX)	**7.2**	**2.6**	**71.1**
Gemma 3n E2B	–	51.4	–	–

Clinical fine-tune of IndicConformer 120M. No public Bengali clinical speech exists, so I built training data: 1,500 template sentences (drugs, doses, tests, symptoms, replies) voiced by IndicF5 cloning 64 real OpenSLR speakers (different TTS model and voices from the test); no test sentence is in the training text. After a quality check, 1,012 clips (81 min) were mixed 1:1 with real speech and the CTC branch trained for 1,500 steps. On the synthetic test, term accuracy rose 51.6 → 70.7 %, with similar gains on seen (50.9 → 69.7) and unseen (44.1 → 61.7) names; on my consultations CER fell

11.1 → 9.4 %, short replies got worse. The pre-set rule (replace the 600M only if consultation CER ≤ 7.2) keeps the 600M. Training and test clinical speech are synthetic, so this is not clinical validation.

Table 8: Task 2 deliverables.

Deliverable	Where
ASR implementation and fine-tuning code	`task2_asr/` (`evaluate_asr.py`, `finetune_whisper.py`, `finetune_indicconformer.py`, `eval_consult.py`)
Sample dataset with reference transcripts	`data/samples/asr/` (10 clips + transcripts); `data/asr/` splits; `data/consultation/`
Baseline and fine-tuned WER / CER	Table 4
Three or more actual errors	Table 5
Before / after of an improvement	Table 6

3 Task 3: Integrated SD–ASR pipeline

3.A Pipeline integration

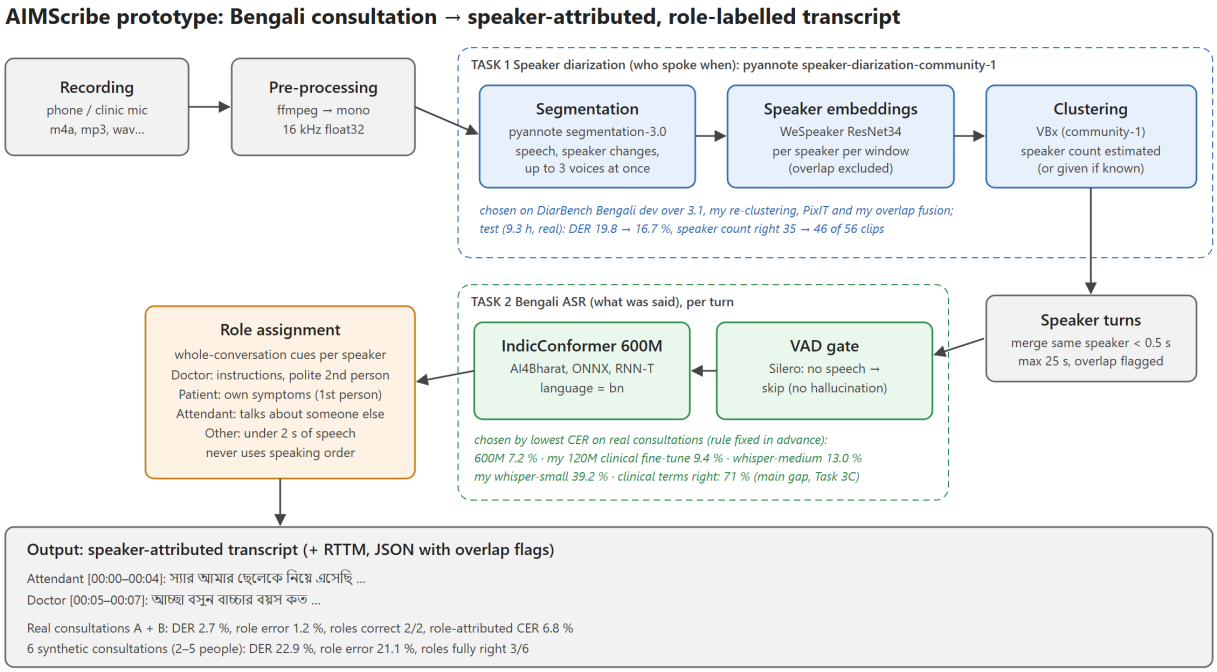


Figure 2: System architecture with the measured result of each stage.

`task3_pipeline/pipeline.py`: audio (any format) is converted to 16 kHz mono; community-1 gives speaker segments; same-speaker segments less than 0.5 s apart are merged into turns of at most 25 s; each turn passes the VAD gate and IndicConformer 600M; roles are assigned per speaker; turns that overlap another speaker are marked. Output: transcript, RTTM and JSON. Unedited output on my consultation B (the attendant speaks first, the child patient never speaks):

Attendant [00:00–00:04]	স্যার আমার ছেলেকে নিয়ে এসেছি ও গতকাল থেকে পাতলা পায়খানা হচ্ছে
Doctor [00:05–00:07]	আচ্ছা বসুন বাচ্চার বয়স কত
Attendant [00:07–00:07]	চার বছর
Doctor [00:08–00:10]	দিনে কয়েকবার পাতলা পায়খানো হচ্ছে
Attendant [00:10–00:15]	প্রায় ছয় সাতবার আর রাতে একবার বমিও করেছে তারপর থেকে
Doctor [00:15–00:15]	জ্বর আছে

Every role and speaker change is right; errors: an inserted word (কয়েকবার), a misspelling (পায়খানো), and later the drug name মেট্রোনিডাজল → ম্যাট্রোনিডাজার.

3.B Clinical speaker roles

Approach (`roles.py`). A role belongs to a speaker, not to a sentence: it is decided from everything that speaker says over the whole visit, never from speaking order.

Role	Cue
Doctor	polite second-person instructions (খাবেন, করাবেন, বসুন), clinical words (টেস্ট, মিলিগ্রাম)
Patient	the speaker's own symptoms (আমার জ্বর, আমার ব্যথা)
Attendant	talks about someone else (ওর, উনার, আমার ছেলে)
Other	less than 2 s of speech

Constraints: one doctor, at most one patient, any number of attendants. Words are matched fuzzily because the input is ASR output.

The cases in the brief. *Patient speaks first:* speaking order is never used. *Several attendants:* allowed; if two speakers both look like the patient, the one with more first-person symptoms keeps Patient and the other becomes Attendant. *Silent patient* (added case): the attendant describes the patient, and no speaker is labelled Patient. *Overlap:* both overlapping turns are kept and marked "(overlap)"; roles are per speaker, so one overlap cannot change a role.

Table 9: Roles per consultation. "Rules only": the role rules applied to the reference speakers and text, i.e. with perfect diarization and ASR. Synthetic: TTS speech, scored once. Real A, B: development data. Roles right = same set of roles as the reference and role error $< 10\,\%$.

Consultation	Speakers (true / found)	DER	Role err.	Roles right	Rules only
S01 patient speaks first	2 / 2	11.1	4.6	✓	✓
S02 silent patient, attendant speaks	2 / 2	10.8	4.5	✓	✓
S05 doctor interrupts (overlap)	2 / 2	6.7	5.7	✓	✓
S03 doctor, patient, 2 attendants, nurse	5 / 2	35.8	41.9	×	✓
S04 3 speakers	3 / 2	36.3	34.4	×	✓
S06 3 speakers, fan noise	3 / 2	33.7	31.9	×	✓
Real A (doctor first)	2 / 2	1.6	0.1	✓	✓
Real B (attendant first, silent patient)	2 / 2	4.1	2.6	✓	✓
Total				**5 / 8**	**8 / 8**

Finding. The role rules are right whenever the speakers are separated correctly (8/8). Every end-to-end failure is a consultation with 3 or more speakers in which diarization merged two people, and a merged speaker cannot carry two roles. With pyannote 3.1 + re-clustering the speakers are separated better on these short synthetic turns (S04 fixed, 4/6), but that front end is worse on 9.3 h of real speech, so the pipeline keeps community-1. Changing the system after seeing these test results would turn the test into tuning data.

Proposed upgrade. (1) Ask the number of people at check-in and pass it to the diarizer (`--num_speakers` exists; on the 10 simulated conversations, giving re-clustering the true count lowers its DER from 13.3 to 6.8 %). (2) Replace keyword counts with a classifier or LLM that scores the four roles from each speaker's full transcript, keeping the same constraints. (3) Use voice cues: a child's voice or very short speech points to Other or Patient. (4) Show low-confidence roles to the doctor for a one-tap confirmation. Evaluate role error and role-set accuracy on held-out real consultations, reported separately for 2-speaker and 3+-speaker visits.

3.C System improvement

Major limitation: clinical vocabulary. Ordinary Bengali is nearly solved (1.5 % CER on the plain synthetic sentences), but even the best model gets about 3 in 10 drug, dose and test names wrong (Table 7), and a wrong drug or dose is a patient-safety risk.

- **Why it occurs:** no public Bengali clinical speech exists (the MedAI corpus is proprietary, DocTalkBN has only YouTube audio with automatic captions, MediBeng repeats 24 sentences), and English drug names spoken in Bengali have no fixed spelling.
- **Component:** the ASR model, plus a post-correction step.
- **Implementation:** (1) record consented real consultations with a spelling guide for drug names; (2) fine-tune the trainable IndicConformer 120M on them with the recipe already tested on synthetic data (51.6 $\rightarrow$ 70.7 % term accuracy); (3) match words against the clinic's drug list and show matches as suggestions for the doctor to confirm, never as silent edits.
- **Evaluation:** drug and dose accuracy on held-out real consultations, split into drugs seen and unseen in training, plus overall CER and the rate of wrong corrections.

Second limitation: rooms with 3 or more speakers (Section 3.B).

Model selection and compatibility

Diarization: community-1, chosen on DiarBench dev and confirmed on the test set (Table 2). **ASR:** IndicConformer 600M, chosen by the pre-set rule on real consultations (Table 7). **Compatibility:** both take 16 kHz mono audio; diarization returns segments in seconds, which become the ASR turns ($\leq$ 25 s); community-1 needs pyannote.audio 4 and runs on the GPU, IndicConformer runs on the CPU through onnxruntime. Speed: about 0.9$\times$ real time on a Colab T4, 2.3$\times$ on the laptop.

Table 10: Task 3 deliverables.

Deliverable	Where
Prototype SD–ASR integration code	`task3_pipeline/pipeline.py`, `roles.py`, `evaluate_pipeline.py`
Sample speaker-attributed transcript	Section 3.A; `results/task3/final_c1/*_transcript.txt`
System architecture diagram	Figure 2; `report/system_diagram.png`
Model selection and compatibility	section above
Improvement proposal	Section 3.C
Presentation	separate file

4 Limitations, hardware and incomplete experiments

- **Hardware:** laptop RTX 3050 (4 GB) cannot fine-tune; all training and large evaluations ran on a free Colab T4, which limited model size and the number of training runs.
- **Small real test:** my consultations are 2.2 min and were used for development; their reference timestamps were drafted from a system output and corrected by ear, so DER on them may favour the system.
- **Synthetic clinical data:** the clinical test, the synthetic consultations and the clinical training data are TTS speech from scripts I wrote; they are cleaner than real speech and are not clinical validation.
- **Role rules** were revised twice after inspecting consultation B, then checked on unseen synthetic consultations.
- **Not completed:** a held-out real consultation (Script C) was not recorded; the 60-clip listening check of the synthetic test audio was not done (no clip was excluded).

5 Own contributions and reproducibility

My work: the consultation scripts, recordings and reference labels; the evaluation protocol and data (DiarBench split with overlap RTTM, simulated conversations, synthetic clinical test and training sets with leakage checks); segment re-clustering; overlap fusion; the whisper-small and IndicConformer-120M fine-tuning recipes; the error analyses and contamination check; the model comparison; the pipeline and role rules; the evaluation code (DER inside and outside overlap, role error, role-attributed WER, cpWER, term accuracy). **Used as they are:** pyannote, Whisper, IndicConformer, Parler-TTS, IndicF5, Silero VAD; public data OpenSLR 53 and Indic DiarBench.

Reproducibility. One Colab notebook per task re-runs the evaluations from public data and compares every number with the reported one: Task 1 24/24, Task 2 27/27, Task 3 19/19 within tolerance. This confirms the numbers, not their generalisation. The default run evaluates the saved fine-tuned models and the saved PixIT output rather than re-training or re-running them. The full submission is on Google Drive (https://drive.google.com/drive/folders/1njMEFzB3lV_t9gzqK0UdIe3RrVoisOUT?usp=sharing). Models and evaluation data are on Hugging Face:

- whisper-small fine-tune: https://huggingface.co/sahilfarib/aimscribe-whisper-small-bn
- IndicConformer-120M clinical fine-tune: https://huggingface.co/sahilfarib/aimscribe-indicconformer-120m-bn-clinical
- evaluation data: https://huggingface.co/datasets/sahilfarib/aimscribe-bengali-eval